

PERIODONTAL LIGAMENT- :Development and Function

DR SHREYA SHETTY
BDS, MDS(PERIODONTICS), FICOI

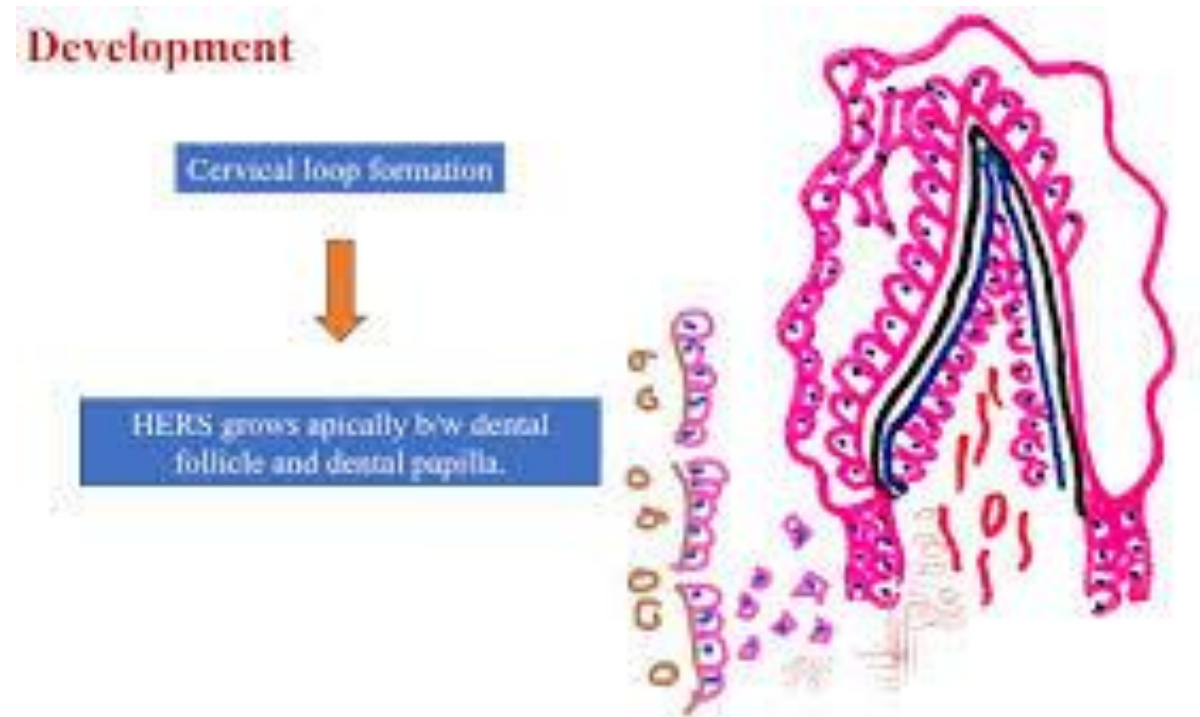
LECTURE LEARNING OUTCOMES

Describe the development of periodontal ligament.

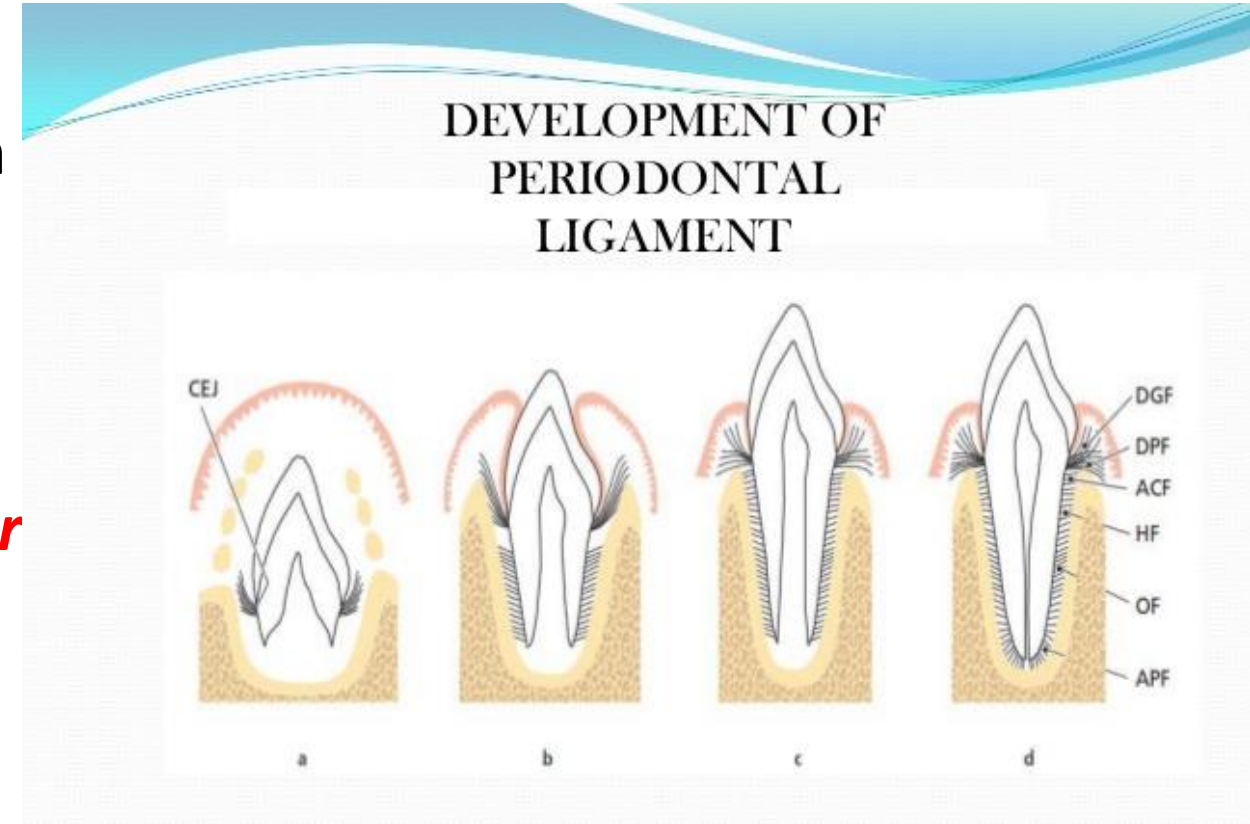
Discuss the role of various components of the PDL in carrying out important functions.

DEVELOPEMENT OF PDL

- The development of the periodontal ligament begins with root formation, prior to tooth eruption.
- The continuous proliferation of the internal and external enamel epithelium forms the cervical loop of the tooth bud.
- This sheath of epithelial cells grows apically, in the form of *Hertwig's epithelial root sheath (HERS)*, between the dental papilla and the dental follicle.

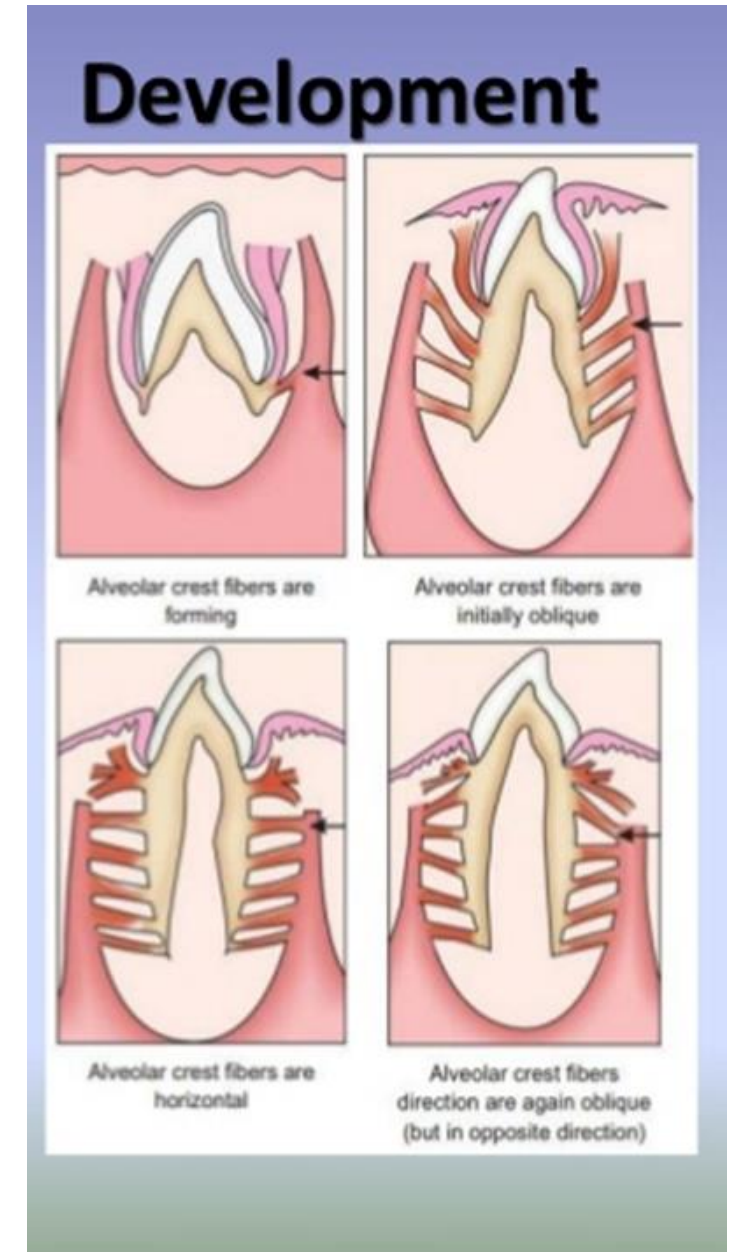


- At this stage, the sheath forms a circumferential structure encompassing dental papilla separating it externally from dental follicle cells.
- The dental follicle cells located between the alveolar bone and the epithelial root sheath are composed of two subpopulations, **mesenchymal cells** of the **dental follicle proper** and the **perifollicular mesenchyme**.
- The developing periodontal ligament and mature periodontal ligament contain undifferentiated stem cells that retain the potential to differentiate into osteoblasts, cementoblasts and fibroblasts.



Development of the principal fibers

- Immediately before tooth eruption, and for sometime thereafter, active fibroblasts adjacent to cementum of the coronal third of the root, appear to become aligned in an oblique direction to the long axis of the tooth.
- Soon thereafter, the **first collagen fiber bundles** of the ligament become discernible. These are the precursors of the **alveolar crest fiber bundle group**.



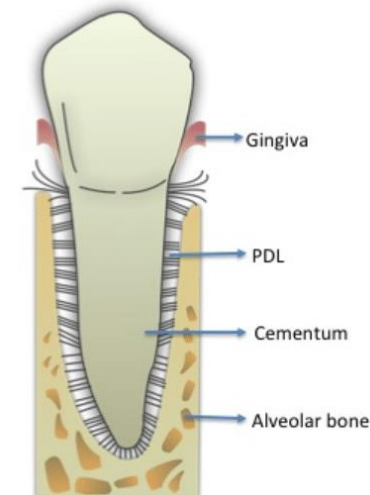
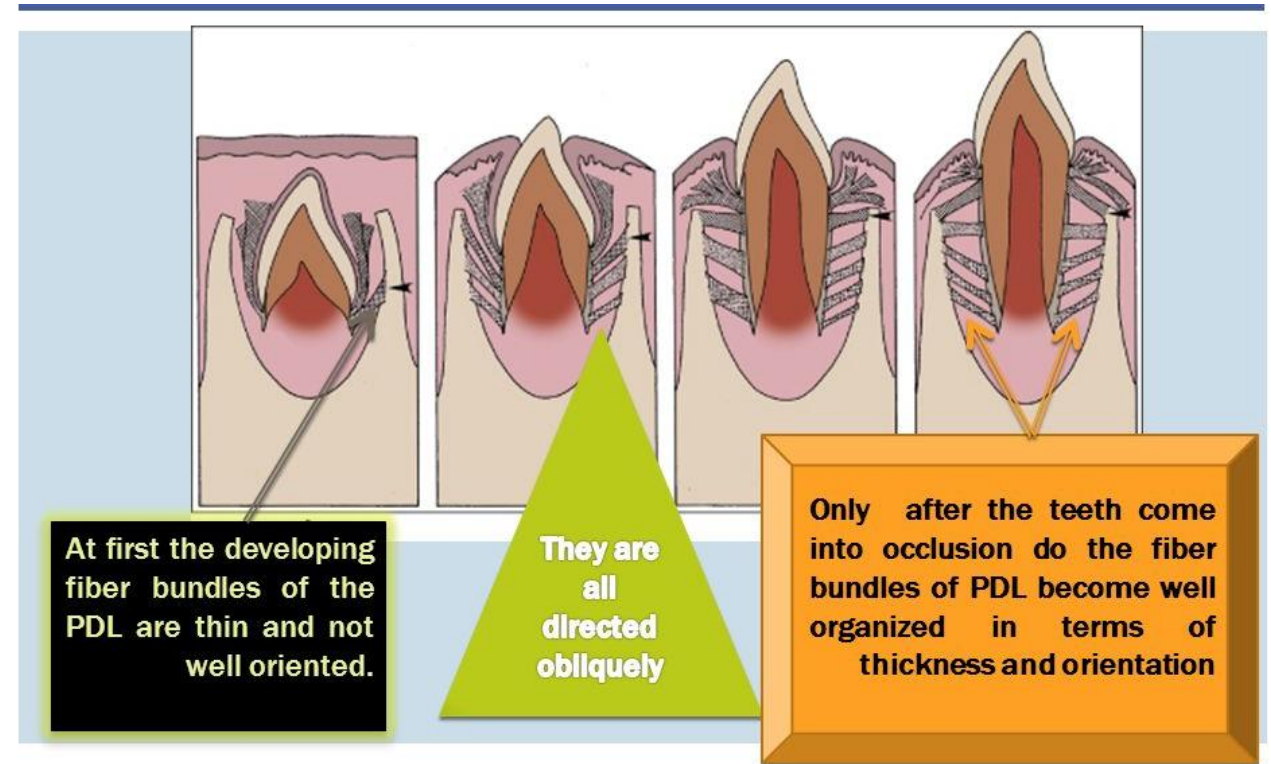
Orban's oral histology and embryology; 13th edition; chapter 8 – periodontal ligament, page 173-174

By the time of **first occlusal contact of the tooth** with its antagonist, the principal fibers around the coronal third of the root, the **horizontal group** are almost completely developed.

The **oblique fibers** in the middle third of the root are still being formed.

As eruption continues, and definite occlusion is established, there is a progressive apical maturation of **oblique fiber bundles**.

With the formation of the **apical fiber group**, the definitive periodontal ligament architecture is established.



FUNCTIONS OF PDL

FUNCTIONS

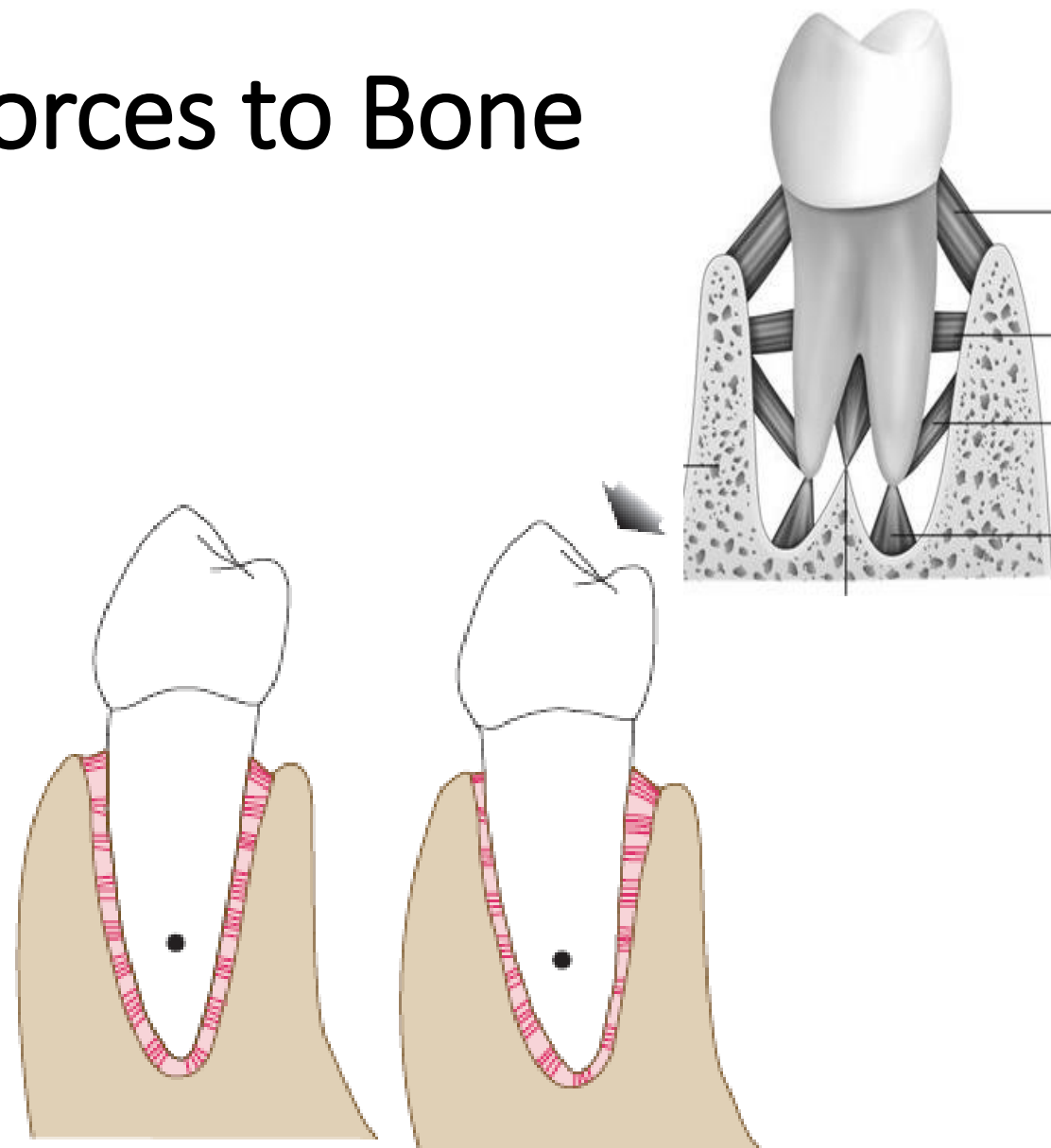
- The periodontal ligament has the following functions:
- Physical
- Supportive
- Homeostatic
- Sensory
- Eruptive
- Nutritive

Physical Functions

- The physical functions of the periodontal ligament entail the following:
 - 1. Provision of a soft-tissue “casing” to protect the vessels and nerves from injury by mechanical forces
 - 2. Transmission of occlusal forces to the bone
 - 3. Attachment of the teeth to the bone
 - 4. Maintenance of the gingival tissues in their proper relationship to the teeth
 - 5. Resistance to the impact of occlusal forces (i.e., shock absorption)

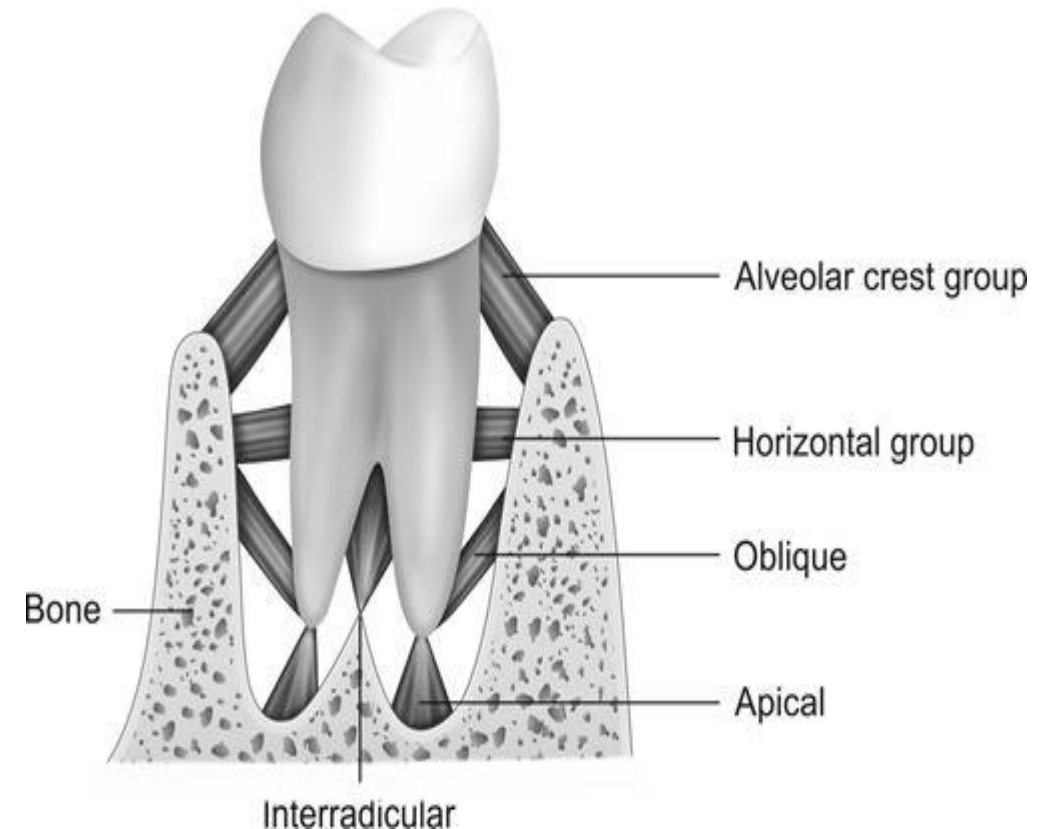
Transmission of Occlusal Forces to Bone

- The arrangement of the principal fibers is similar to that of a suspension bridge or a hammock.
- When an axial force is applied to a tooth, a tendency toward displacement of the root into the alveolus occurs. The oblique fibers alter their wavy, untensed pattern, assume their full length, and sustain the major part of the axial force.
- When a horizontal or tipping force is applied, two phases of tooth movement occur. The first is within the confines of the periodontal ligament, and the second produces a displacement of the facial and lingual bony plates.
- The tooth rotates about an axis that may change as the force is increased. The apical portion of the root moves in a direction that is opposite to the coronal portion.
- In areas of tension, the principal fiber bundles are taut rather than wavy. In areas of pressure, the fibers are compressed, the tooth is displaced, and a corresponding distortion of bone exists in the direction of root movement.



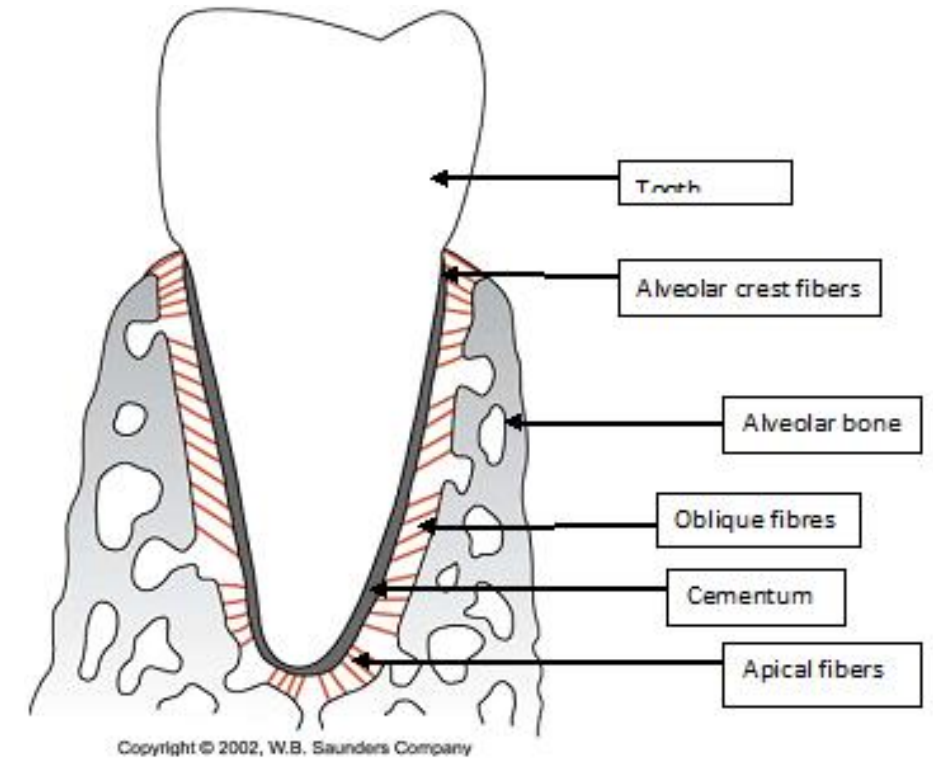
Resistance to Impact of Occlusal Forces (Shock Absorption)

- Two theories pertaining to the mechanism of tooth support have been considered: **the tensional theory** and the **viscoelastic system theory**.
- The **tensional theory** of tooth support states that
 - the principal fibers of the periodontal ligament are the major factor in supporting the tooth and transmitting forces to the bone.
 - When a force is applied to the crown, the principal fibers first unfold and straighten, and they then transmit forces to the alveolar bone, thereby causing an elastic deformation of the bony socket.
 - Finally, when the alveolar bone has reached its limit, the load is transmitted to the basal bone. Many investigators find this theory insufficient to explain available experimental evidence.



The **viscoelastic system** theory states that the displacement of the tooth is largely controlled by **fluid movements**, with fibers having only a secondary role.

- When forces are transmitted to the tooth, the **extracellular fluid** passes from the periodontal ligament into the marrow spaces of the bone through the foramina in the cribriform plate.
- After the depletion of tissue fluids, the fiber bundles absorb the slack and tighten. This leads to a blood vessel stenosis.
- Arterial back pressure causes ballooning of the vessels and passage of the blood ultrafiltrates into the tissues, thereby replenishing the tissue fluids



Formative and Remodeling Function

- Periodontal ligament and alveolar bone cells are exposed to **physical forces in response to mastication, parafunction, speech, and orthodontic tooth movement.**
- Cells of the periodontal ligament participate in the **formation and resorption of cementum and bone**, which occur during physiologic tooth movement, during the accommodation of the periodontium to occlusal forces, and during the repair of injuries.
- The periodontal ligament is constantly undergoing remodelling. Old cells and fibers are broken down and replaced by new ones, and mitotic activity can be observed in the fibroblasts and the endothelial cells.

Nutritional and Sensory Functions

- The periodontal ligament **supplies nutrients to the cementum, bone, and gingiva by way of the blood vessels, and it also provides lymphatic drainage.**
- In relation to other ligaments and tendons, the periodontal ligament is highly vascularized tissue.
- The periodontal ligament is abundantly supplied with sensory nerve fibers that are capable of transmitting **tactile, pressure, and pain sensations via the trigeminal pathways.**

Regulation of Periodontal Ligament Width

- Some of the most **interesting features** of the periodontal ligament in animals are its adaptability to rapidly changing applied force and its **capacity to maintain its width at constant dimensions throughout its lifetime.**
- These are important measures of **periodontal ligament homeostasis** that regulate the formation of bone, cementum, and periodontal ligament fibers.
- In addition, the ability of periodontal ligament cells to synthesize and secrete a wide range of regulatory molecules is an essential component of tissue remodeling and periodontal ligament homeostasis.

AGE CHANGES

- decreased numbers of fibroblasts
- decreased organic matrix production, decreased epithelial cell rests, and increased amounts of elastic fiber.
- Cell proliferation decreased with age, thereby suggesting an impairment of repair potential.

Newman and Carranza's clinical periodontology – 13th edition;
chapter 4 – Aging and the periodontium; page 52

THANK YOU